

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B4. SUB-ATOMIC PHYSICS

TRINITY TERM 2017

Wednesday, 7 June, 2.30 pm – 4.30 pm

Candidates are strongly advised to use the first 10 minutes
to read the whole paper before starting writing.

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

*The numbers in the margin indicate the weight that the Examiners expect to
assign to each part of the question.*

Do NOT turn over until told that you may do so.

1. Consider the simplest bound nuclear states, the deuteron (pn , $B = 2.2$ MeV) and the triton (pnn , $B = 8.5$ MeV), and the nearly bound dineutron (nn , $B \approx 0$), where B is the binding energy.

(a) These states can be used to study the asymmetry and pairing terms in the semi-empirical mass formula.

- (i) Discuss how the spins of the proton and the neutron affect the binding and contribute to the asymmetry term in the semi-empirical mass formula.
- (ii) The Fermi energy can be calculated for a nuclear radius of 1.2 fm to be $E_f = 32$ MeV in the lowest energy state. The average kinetic energy of a nucleon is $3E_f/5$. Calculate the average total kinetic energy of nucleons in the deuteron, triton, and dineutron. Determine the potential energy for each state.
- (iii) How does the pairing energy δ contribute to the binding energy for each of the states?
- (iv) In the case of a $p - n$ bound state the spins are aligned in a spin-1 state. Would an $n - n$ bound state have aligned or anti-aligned spins? The spin-coupling energy contributes $a\langle\vec{s}_1 \cdot \vec{s}_2\rangle$ to the binding energy, which is $a/4$ for a spin-1 state and $-3a/4$ for a spin-0 state (a is a constant with units of energy). Write down this contribution to the deuteron, triton, and dineutron states (triton has spin-1/2).
- (v) Take the binding energy of the strong force to be a constant B_s for each bond between hadrons. Considering the number of available bonds in the states, write an expression for the strong binding energy for each of the three states in terms of B_s .
- (vi) Take the sum of the above three terms to be the negative total potential energy of each state. Use the three equations to solve for the three unknowns a , B_s and δ .
- (vii) A fit to the complete set of nucleons suggests a parameterization of the pairing term of $\delta \approx 12/A^{1/2}$. What is the ratio of this parameterization to the δ calculated above using only the lowest lying states?
- (viii) In this calculation we have replaced the volume and surface terms of the liquid drop model with an explicit accounting of the hadronic bonds. What other term in the liquid drop model does not appear in this calculation, and why not?

[20]

(b) Triton has important practical applications.

- (i) Triton decays with a half-life of 12.3 years. What particles does it decay to?
- (ii) The Q value of the triton decay is 18.6 keV. How can it be used to measure the electron-neutrino mass?
- (iii) Triton is involved in a promising reaction for a fusion reactor. What is this reaction? Give two reasons why the reaction is promising.

[5]

2. Nuclear structure and size can be studied via the Coulomb force.

(a) Approximate the nucleus as a sphere with a uniform charge distribution with radius r .

- (i) In this case the Coulomb energy is $V = 3Z^2e^2/(20\pi\epsilon_0r)$, where Z is the nuclear charge, $e = 1.6 \times 10^{-19}$ C, and $\epsilon_0 = 8.9 \times 10^{-12}$ C/(V·m). Writing the energy as $V = \alpha_C Z^2/r$, calculate α_C in units of MeV·fm. Take the nuclear radius to be $r = 1.2A^{1/3}$ fm and determine V for the proton. Compare this to the empirical value $V = 0.7$ MeV obtained from a fit to the masses of bound nuclei and comment on the accuracy of this charge distribution for the binding energy calculation.
- (ii) For a given A the Coulomb energy would tend to reduce the number of protons in the nucleus. What counters this tendency?
- (iii) ${}^{11}_6\text{C}$ (mass $11.0081u$) and ${}^{11}_5\text{B}$ (mass $11.0066u$) are “mirror nuclei” whose binding energy only differs by the Coulomb term. Calculate the difference in Coulomb terms between these nuclei (the proton mass is $1.0073u$ and the neutron mass is $1.0087u$; take $u = 931.5$ MeV/ c^2). Use this difference to estimate the radius of these nuclei, assuming a common radius and a uniform charge distribution in a sphere. Compare this radius to the parameterization $r = 1.2A^{1/3}$ fm and give a reason for the difference.

[13]

(b) The nuclear charge distribution can be refined by studying electron-nucleon scattering. By measuring the form factor as a function of momentum transfer the charge distribution can be inferred. In units where $\hbar = 1$, the form factor for Coulomb scattering is $F(\vec{q}) = \int e^{-i\vec{r}\cdot\vec{q}}\rho(\vec{r})d^3\vec{r}$.

- (i) Show that the form factor of a spherically symmetric distribution is $F(q) = \frac{4\pi}{q} \int r \sin(qr)\rho(r)dr$.
- (ii) Derive a formula for the form factor for a uniform charge distribution with density ρ_0 within a radius a .
- (iii) Measurements of electron-nucleon scattering show that the nuclear charge is well modelled by a Saxon-Woods distribution, $\rho(r) = \rho_0/(1 + e^{(r-a)/d})$, where a is a measure of the radius of the nucleus and d is a measure of the thickness of its surface ($d \ll a$). The calculation of the form factor for this distribution is complicated but can be performed in an approximated distribution where $\rho = \rho_0$ for $r < a-d$, $\rho = \rho_0[1/2 - (r-a)/2d]$ for r between $a-d$ and $a+d$, and $\rho = 0$ for $r > a+d$. Sketch this approximation along with the Saxon-Woods distribution. Integrate the form factor using the approximation separately for $r < a-d$ and $a-d < r < a+d$; the total form factor is the sum.

[12]

3. The strong interaction can be investigated in detail by studying meson masses and decays.

(a) The energy due to angular momentum contributes significantly to hadron masses.

- (i) Consider a meson composed of a quark and antiquark. The spin-coupling energy can be expressed as $E = a\vec{S}_q \cdot \vec{S}_{\bar{q}}/(m_q m_{\bar{q}})$. Given the mass of the charged pion ($140 \text{ MeV}/c^2$) and the constituent masses of the up and down quarks ($m_u = m_d = 340 \text{ MeV}/c^2$), calculate a . Use this value to estimate the mass of the charged ρ meson and find the ratio of the estimate to the experimental value of $780 \text{ MeV}/c^2$.
- (ii) For states with non-zero angular momentum we can calculate the mass using an expression for the potential energy: $V = -4\alpha_s \hbar c/(3r) + kr$. Give a physical explanation for each term and sketch the potential using $\alpha_s = 0.2$ and $k = 1 \text{ GeV}/\text{fm}$.
- (iii) Determine the relationship between the energy (from the linear term in the potential) and the angular momentum (from the relativistic rotation of the energy string connecting the quark and antiquark). Note that $\int_0^1 dx/\sqrt{1-x^2} = \pi/2$ and $\int_0^1 x^2 dx/\sqrt{1-x^2} = \pi/4$. Calculate the contribution of this potential to the mass of the $l = 1$ state $b_1(1235)$ (a $u\bar{d}$ meson) and take its ratio with respect to the $1100 \text{ MeV}/c^2$ difference between the mass of the b_1 meson to the mass of the charged π meson. Take $\hbar c = 0.2 \text{ GeV}\cdot\text{fm}$ and $k = 1 \text{ GeV}/\text{fm}$.

[18]

(b) Consider the J/ψ meson decay.

- (i) Three possible J/ψ meson decays are to $\pi^0\pi^+\pi^-$, $\omega^0\pi^+\pi^-$, and $\mu^+\mu^-$. Draw the leading order Feynman diagrams representing these decays. Explain why the branching fraction to $\mu^+\mu^-$ is similar to $\pi^0\pi^+\pi^-$, and why the branching fraction to $\omega^0\pi^+\pi^-$ is much lower than to $\pi^0\pi^+\pi^-$.
- (ii) The J/ψ meson can decay through an intermediate $b_1^+\pi^-$ state, with the b_1^+ subsequently decaying to $\omega^0\pi^+$. Draw a possible Feynman diagram for this process.

[7]

4. The weak force derives its name from its strength relative to the other forces at typical terrestrial energy scales. It nonetheless has important applications to energy production on earth and in the sun.

- (a) The relative weakness of the weak force can be investigated by comparing it to the electromagnetic force at the nuclear energy scale.
- (i) The potential due to the exchange of a massive W or Z boson can be represented by the Yukawa potential, $V(r) = V_0 f(r)$. Give the form of $f(r)$.
 - (ii) Consider a beam of 1 GeV muons incident on a stationary target with a detector to observe outgoing electrons. What are the leading order Feynman diagrams contributing to this process? Assume that the fraction of decaying muons is negligible.
 - (iii) In the first order Born approximation the wavefunction for a particle incident on a radial potential is

$$\psi(\vec{r}) = e^{i\vec{k}\cdot\vec{r}} - 2me^{ikr}/(\hbar^2 qr) \int_0^\infty r' V(r') \sin(qr') dr',$$

where $q^2 = 2k^2(1 - \cos \Theta)$ and m is the mass of the incoming particle and r is the radius from the potential source. Use this to derive the differential cross section $d\sigma/d\Omega$ in terms of k and Θ , where k is the incoming momentum.

- (iv) Consider the detector to cover the range $-0.5 < \cos \Theta < 0.5$ and the entire azimuthal angle. Integrate to find a formula for the total observed cross section.
- (v) Using $V_0 = g_W^2/(32 \times 4\pi)$ and $g_W = 0.6$ for the case of the massive exchanged boson, find the ratio of this cross section to the cross section with an exchanged photon. Is the total observed cross section the sum of these two?

[17]

- (b) The sun is a source of neutrinos that can be studied on earth.
- (i) What fusion reaction produces deuterons in the sun? Give an order of magnitude for the energy of the neutrinos produced in this reaction.
 - (ii) A detector of heavy water is used to measure the following reactions produced by neutrinos from the sun: $\nu + d \rightarrow p + p + e^- - 1.4$ MeV, $\nu + d \rightarrow p + n + \nu_x - 2.2$ MeV, and $\nu + e^- \rightarrow \nu + e^-$. What neutrino flavour(s) can produce each reaction? Draw the weak-boson exchange diagrams for each process.

[8]